

Origin of Structural Variations in Amorphous SiO₂ Generated by Melt-Quench Simulations

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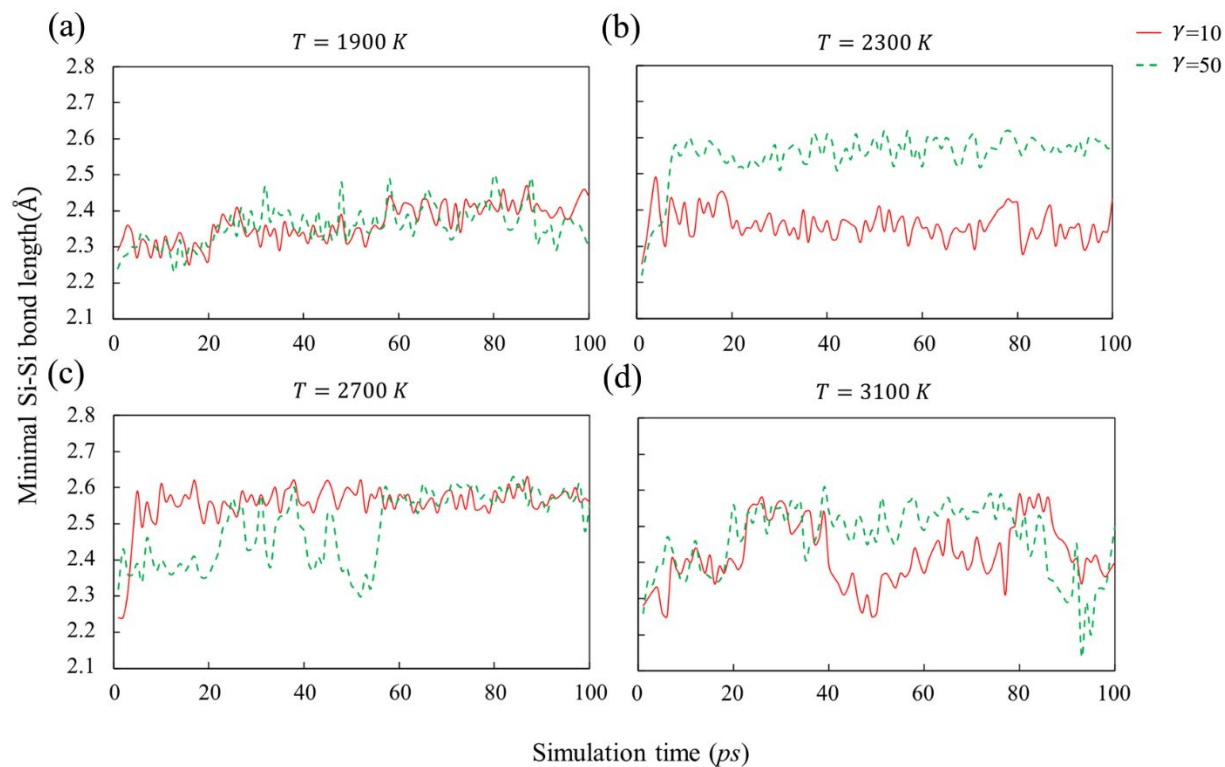


Figure S1. Variation of the minimal Si-Si bond length during melting at different temperatures for system II. Solid line corresponds to a Langevin thermostat friction coefficient of 10 ps^{-1} , while the dashed line corresponds to a friction coefficient of 50 ps^{-1} .

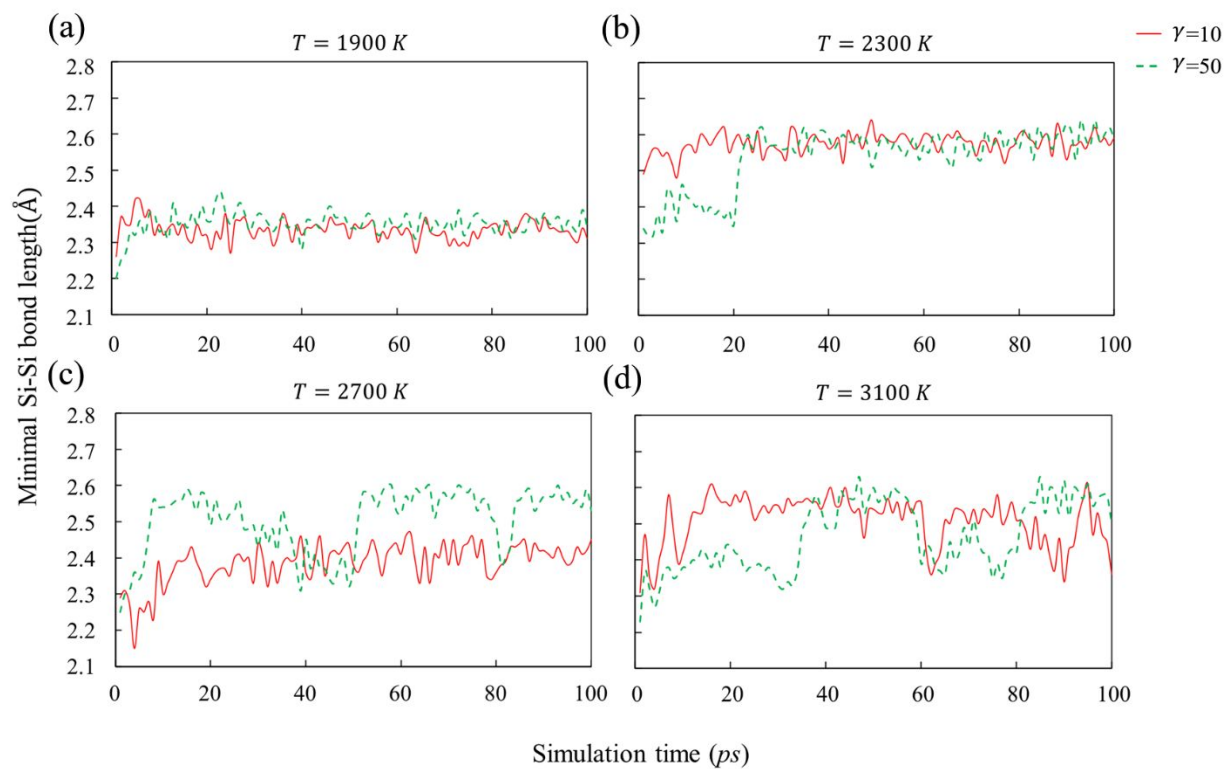


Figure S2. Variation of the minimal Si-Si bond length during melting at different temperatures for system III. Solid line corresponds to a Langevin thermostat friction coefficient of 10 ps^{-1} , while the dashed line corresponds to a friction coefficient of 50 ps^{-1} .

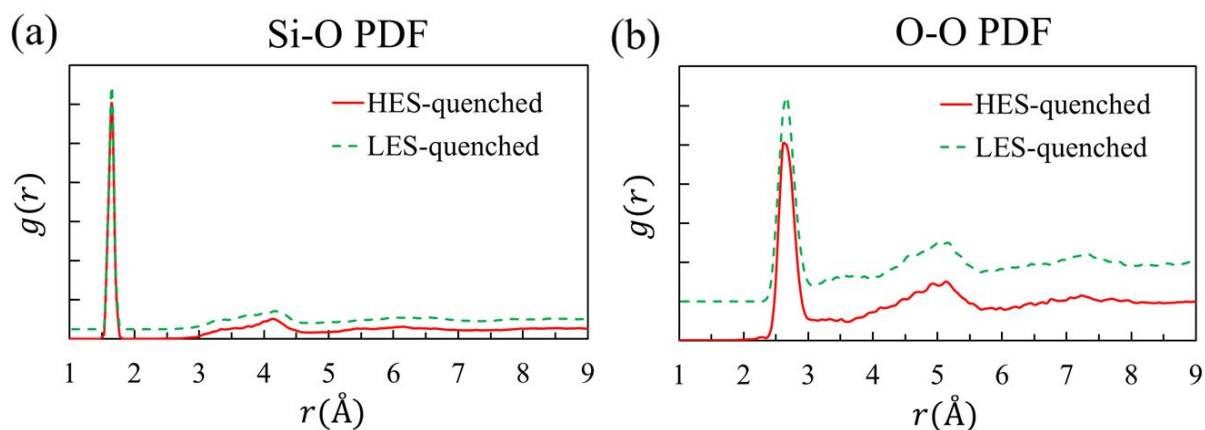


Figure S3. (a) Si-O pair distribution functions for one representative structure quenched from HES and one from LES. (b) O-O pair distribution functions for the same two structures.

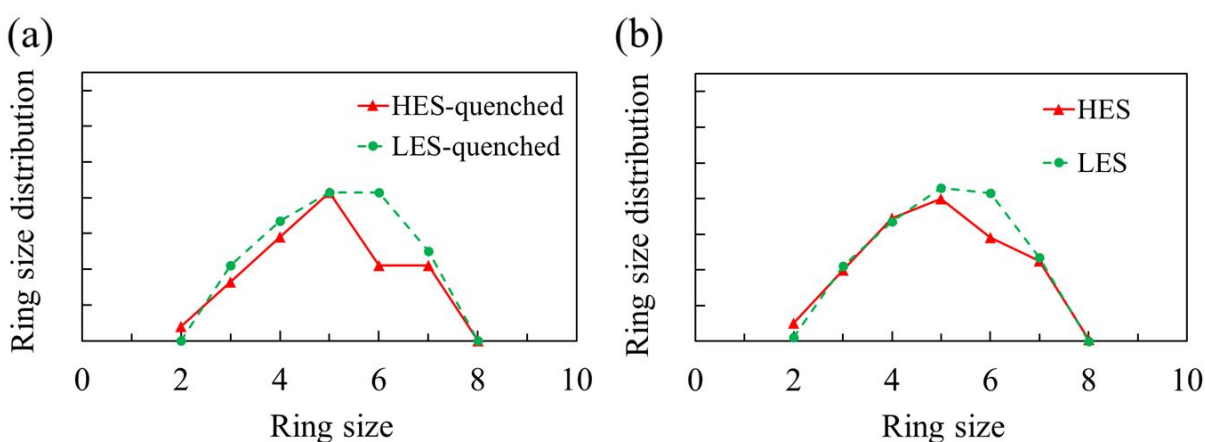


Figure S4. (a) The ring size distribution for one representative structure **quenched** from HES and one from LES. (b) The ring size distribution before quenching for system **I** at 2300 K. The distribution was calculated before the transition, which is the HES state, and after the transition, which is the LES state.

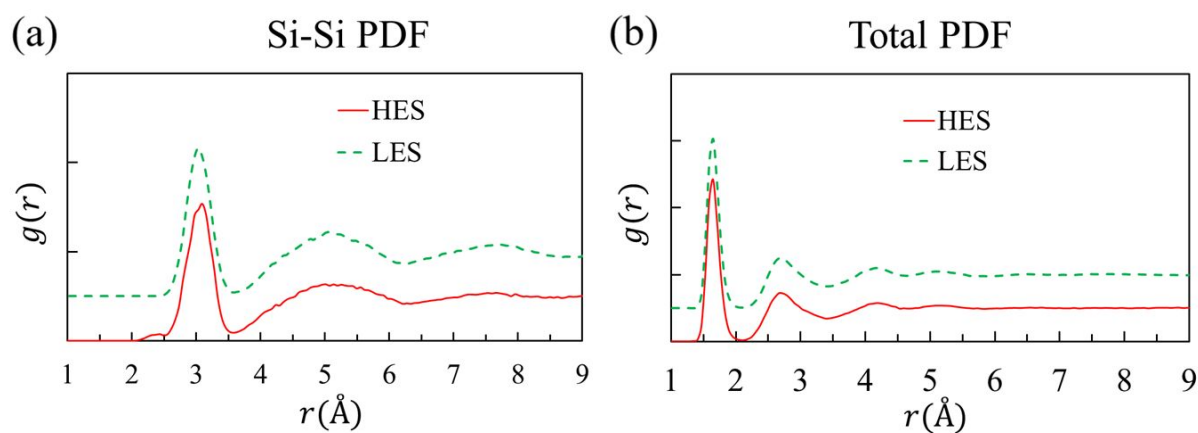


Figure S5. (a) Si-Si pair distribution function for system I at 2300 K in the HES and the LES. (b) The total pair distribution function for system I at 2300K in the HES and LES.

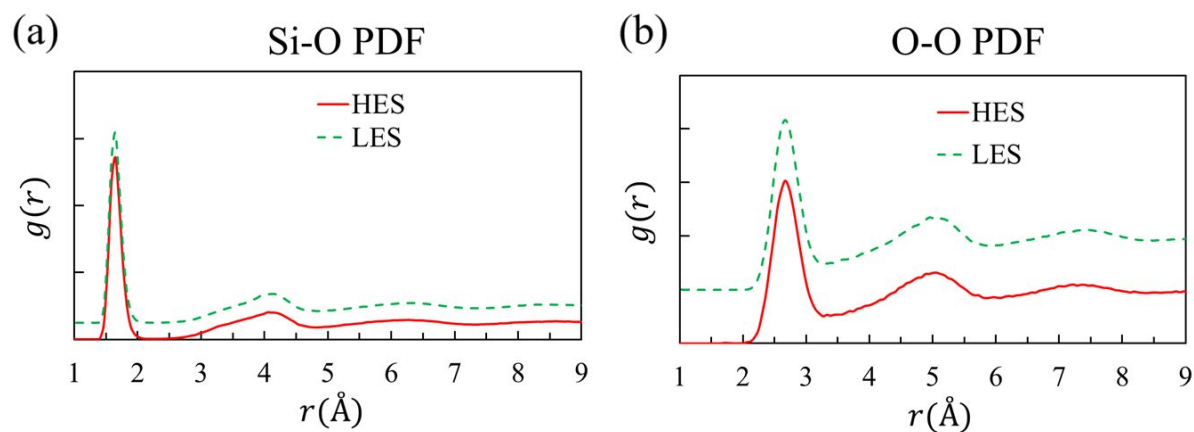


Figure S6. (a) Si-O pair distribution function for system I at 2300 K in the HES and the LES. (b) O-O pair distribution function for system I at 2300K in the HES and LES.

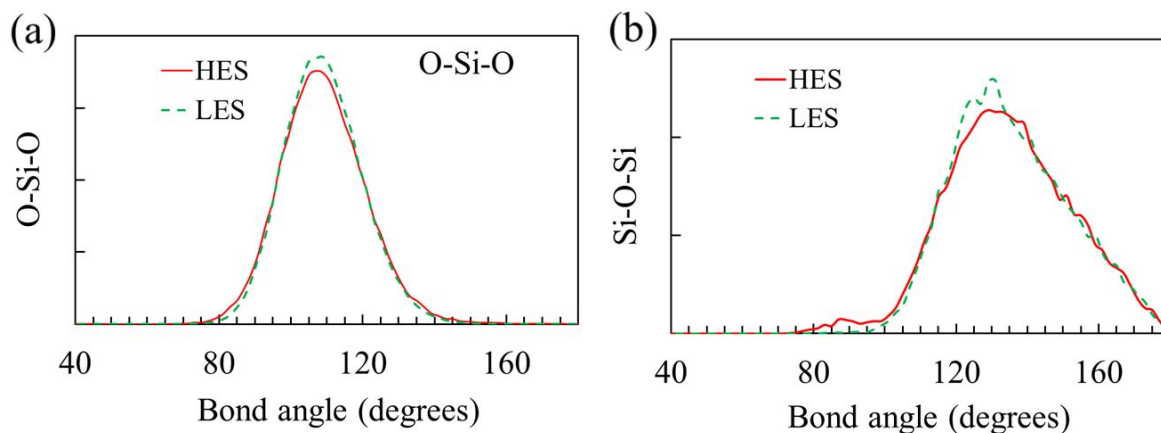


Figure S7. (a) O-Si-O bond angle distribution for system I at 2300 K in the HES and the LES. (b) Si-O-Si bond angle distribution for system I at 2300K in the HES and LES.

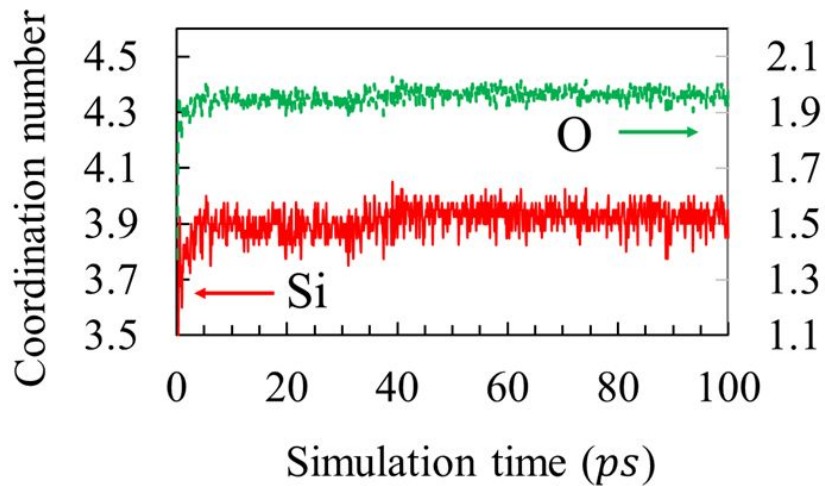


Figure S8. Variation of the Si and O coordination number for system I at 2300 K. A slight increase in the Si and O coordination number is observed at around 40 *ps*.